

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## 4<sup>th</sup> Semester of B. Pharm. Examination

University Theory Examination May 2016

### PH222 Phytochemistry-1

Date: 13.05.16 , Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

#### Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

#### SECTION - I

- |            |                                                                                                                 |           |
|------------|-----------------------------------------------------------------------------------------------------------------|-----------|
| <b>Q 1</b> | Attempt all questions. Each question is of one mark.                                                            | <b>20</b> |
| 1.         | Stahl's method is used for isolation of<br>[A] Digoxin<br>[B] Diosgenin<br>[C] Sennosides<br>[D] Morphine       |           |
| 2.         | Kedde's test is used to identify<br>[A] Diosgenin<br>[B] Digoxin<br>[C] Sennosides<br>[D] Reserpine             |           |
| 3.         | Which of the following is a dimer?<br>[A] Reserpine<br>[B] Atropine<br>[C] Sennoside B<br>[D] Caffeine          |           |
| 4.         | Which of the following is a pseudo alkaloid?<br>[A] Vincristine<br>[B] Morphine<br>[C] Caffeine<br>[D] Atropine |           |
| 5.         | One of following is not Glycoside<br>[A] Diosgenin<br>[B] Digoxin<br>[C] Rutin<br>[D] Morphine                  |           |
| 6.         | Alseroxylon is formed during isolation of<br>[A] Atropine<br>[B] Morphine<br>[C] Reserpine<br>[D] Vincristine   |           |
| 7.         | Which of the following is an ester?<br>[A] Sennoside A<br>[B] Reserpine<br>[C] Atropine<br>[D] Rutin            |           |

8. Quinidine is the \_\_\_\_\_ of Quinine  
[A] Isomer  
[B] Substitute  
[C] Semisynthetic derivative  
[D] Bioequivalent
9. Rubanol is the ring structure in  
[A] Morphine  
[B] Reserpine  
[C] Quinine  
[D] Glycyrrhizin
10. How will you convert alkaloidal salt to free base?  
[A] Esterification  
[B] Acidification  
[C] Common Ion Effect  
[D] Basification
11. Quinine may be isolated from crude fraction using,  
[A] Sublimation  
[B] Crystallization  
[C] Fractional Precipitation  
[D] Selective solvent extraction
12. Which of the following is a xanthine compound?  
[A] Ergotamine  
[B] Caffeine  
[C] Morphine  
[D] Reserpine
13. Which of the following is used in treatment of migraine?  
[A] Caffeine  
[B] Reserpine  
[C] Rutin  
[D] Ergotamine
14. Which of the following is used as Analgesic?  
[A] Digoxin  
[B] Diosgenin  
[C] Ergometrine  
[D] Morphine
15. Which of the following is used as anticancer agent?  
[A] Ephedrine  
[B] Digoxin  
[C] Vincristine  
[D] Ergotamine
16. An important steroidal precursor, 16-DPA, is synthesized from  
[A] Digoxin  
[B] Diosgenin  
[C] Glycyrrhizin  
[D] None of above
17. Digoxin is obtained from  
[A] Digitalis purpurea  
[B] Digitalis lanata  
[C] Both of above  
[D] None of above
18. One of following contains anthraquinone moiety  
[A] Digoxin  
[B] Rutin  
[C] Sennoside  
[D] Dioscine

19. Ergotamine exhibits \_\_\_\_\_  
 [A] Diastereomerism  
 [B] Enantiomerism  
 [C] Meso structure  
 [D] Epimerism
20. Glycosidic structure must possess \_\_\_\_\_  
 [A] Aglycone  
 [B] Glycone  
 [C] Hemiacetal Linkage  
 [D] All of Above

### SECTION – II

- Q 2** Attempt any **FOUR** of the following
- Prove :
- |          |                                                                            |           |
|----------|----------------------------------------------------------------------------|-----------|
| <b>A</b> | 1. Quinine contains Quinoline Moiety                                       | <b>05</b> |
|          | 2. Morphine contains both phenolic group and alcoholic group               |           |
| <b>B</b> | Describe chemistry of Cardiac Glycosides obtained from plants              | <b>05</b> |
| <b>C</b> | Write a detailed note on flavonoidal glycoside - Rutin                     | <b>05</b> |
| <b>D</b> | Write detailed note on Hofmann exhaustive methylation and its applications | <b>05</b> |
| <b>E</b> | Prepare a note on Vinca Alkaloids                                          | <b>05</b> |
| <b>F</b> | Write in details: Emde Degradation and its application                     | <b>05</b> |

### SECTION – III

- Q 3** Attempt any **FOUR** of the following
- |          |                                                                    |           |
|----------|--------------------------------------------------------------------|-----------|
| <b>A</b> | Describe shikimic acid pathway.                                    | <b>05</b> |
| <b>B</b> | Write important steps involved in biosynthesis of Quinine          | <b>05</b> |
| <b>C</b> | Write a note on distribution, chemistry and isolation of Quinine.  | <b>05</b> |
| <b>D</b> | Write a note on distribution, isolation and use of any one lignan. | <b>05</b> |
| <b>E</b> | Write method proposed for isolation of Sennosides.                 | <b>05</b> |
| <b>F</b> | Write a note on distribution, isolation and uses of Reserpine.     | <b>05</b> |
- Q 4** Attempt any **FOUR** of the following
- |          |                                                                       |           |
|----------|-----------------------------------------------------------------------|-----------|
| <b>A</b> | Prepare a note on biosynthesis of Ephedrine                           | <b>05</b> |
| <b>B</b> | Prove Chemical Structure of Ephedrine using Chemical Approach         | <b>05</b> |
| <b>C</b> | Write a note on stereoisomerism in Natural Products giving examples.  | <b>05</b> |
| <b>D</b> | Describe isolation methods of Diosgenin.                              | <b>05</b> |
| <b>E</b> | Write a note on distribution, isolation and pharmacology of Morphine. | <b>05</b> |
| <b>F</b> | Write important steps involved in biogenesis of Morphine              | <b>05</b> |